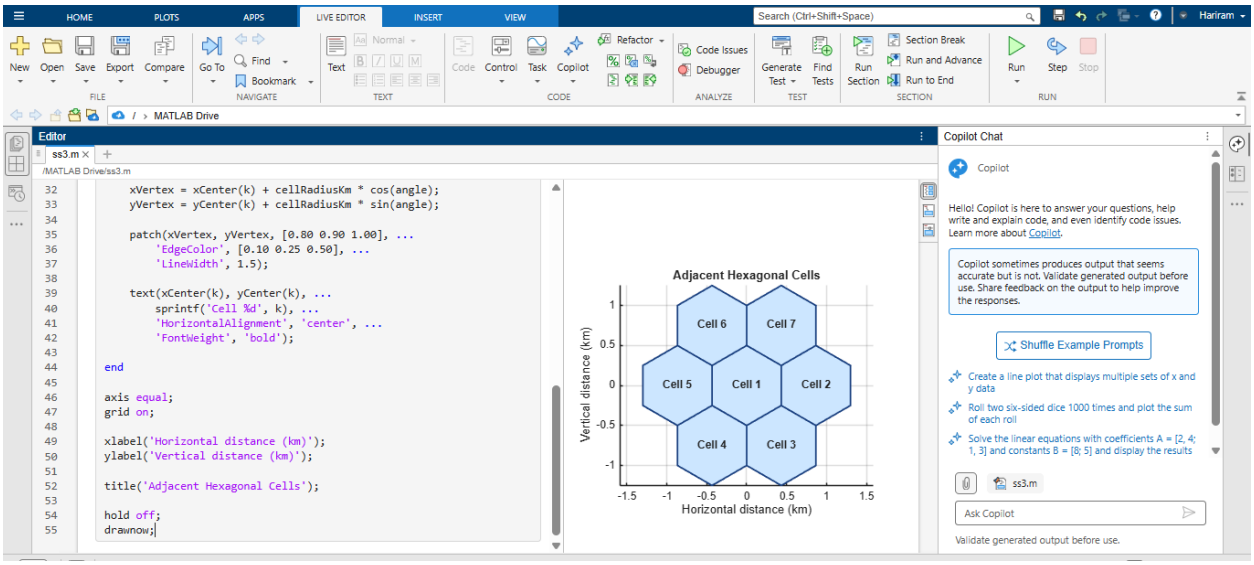
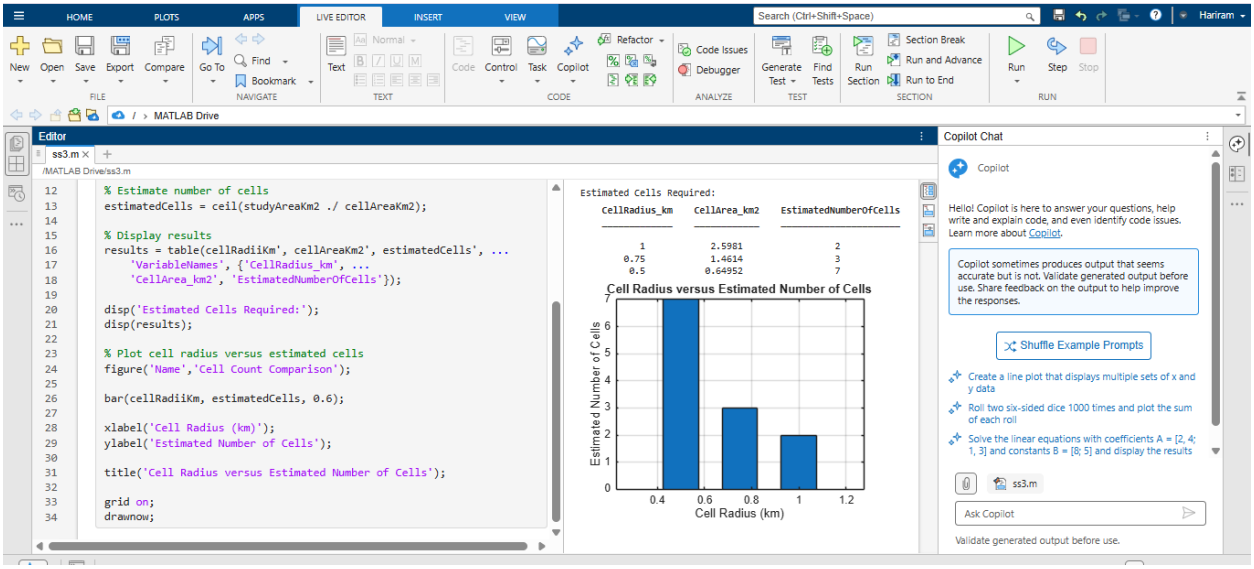
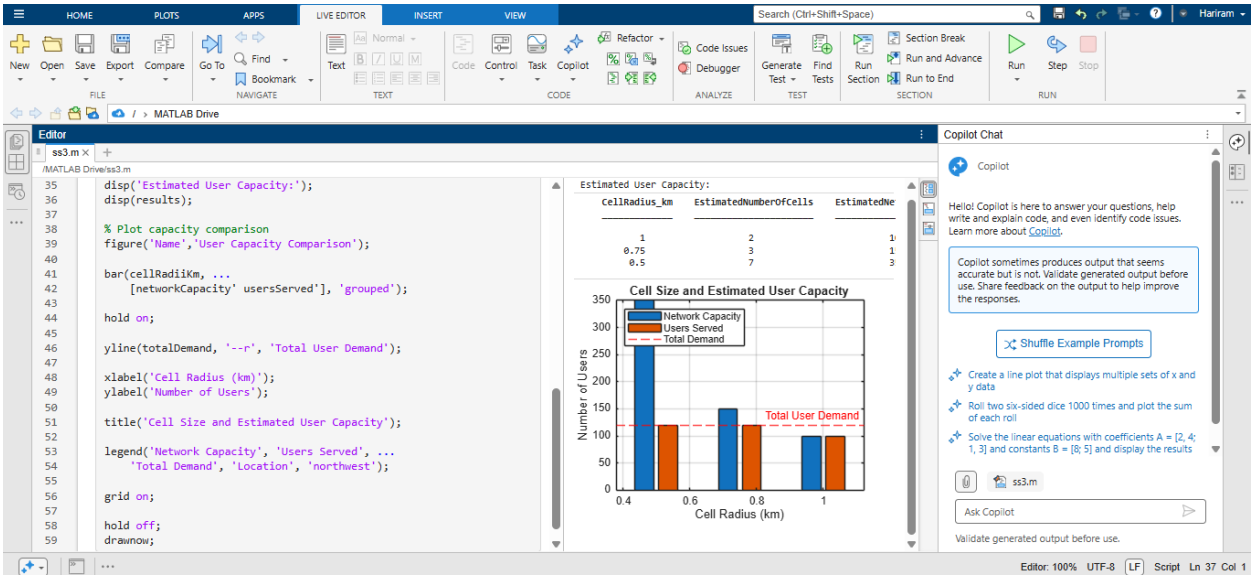
EXP 2(i)


xVertex = xCenter(k) + cellRadiusKm * cos(angle);
yVertex = yCenter(k) + cellRadiusKm * sin(angle);
patch(xVertex, yVertex, [0.80 0.90 1.00], ...
'EdgeColor', [0.10 0.25 0.50], ...
'LineWidth', 1.5);
text(xCenter(k), yCenter(k), ...
sprintf('Cell %d', k), ...
'HorizontalAlignment', 'center', ...
'FontWeight', 'bold');
end
axis equal;
grid on;
xlabel('Horizontal distance (km)');
ylabel('Vertical distance (km)');
title('Adjacent Hexagonal Cells');
hold off;
drawnow;
Adjacent Hexagonal Cells
Cell 6
Cell 7
Cell 5
Cell 1
Cell 2
Cell 4
Cell 3
Vertical distance (km)
Horizontal distance (km)

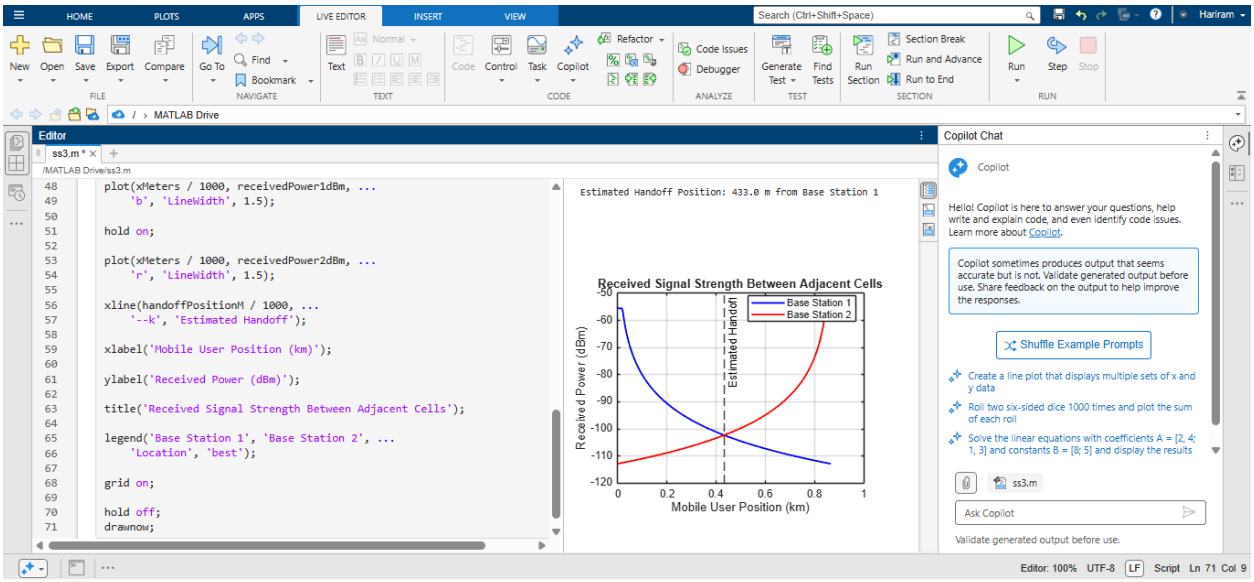
(ii)


% Estimate number of cells
estimatedCells = ceil(studyAreaKm2 ./ cellAreaKm2);
% Display results
results = table(cellRadiiKm', cellAreaKm2', estimatedCells', ...
'VariableNames', {'CellRadius_km', ...
'CellArea_km2', 'EstimatedNumberOfCells'});
disp('Estimated Cells Required:');
disp(results);
% Plot cell radius versus estimated cells
figure('Name','Cell Count Comparison');
bar(cellRadiiKm, estimatedCells, 0.6);
xlabel('Cell Radius (km)');
ylabel('Estimated Number of Cells');
title('Cell Radius versus Estimated Number of Cells');
grid on;
drawnow;
Estimated Cells Required:
CellRadius_km CellArea_km2 EstimatedNumberOfCells
1 2.5981 2
0.75 1.4614 3
0.5 0.64952 7
Cell Radius versus Estimated Number of Cells
Estimated Number of Cells
Cell Radius (km)

(iii)


Editor
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disp('Estimated User Capacity:');
disp(results);
% Plot capacity comparison
figure('Name','User Capacity Comparison');
bar(cellRadiiKm, ...
[networkCapacity' usersServed'], 'grouped');
hold on;
yline(totalDemand, '--r', 'Total User Demand');
xlabel('Cell Radius (km)');
ylabel('Number of Users');
title('Cell Size and Estimated User Capacity');
legend('Network Capacity', 'Users Served', ...
'Total Demand', 'Location', 'northwest');
grid on;
hold off;
drawnow;
Estimated User Capacity:
CellRadius_km
EstimatedNumberOfCells
EstimatedNe
1
2
0.75
3
0.5
7
Cell Size and Estimated User Capacity
Network Capacity
Users Served
Total Demand
Total User Demand
Number of Users
Cell Radius (km)
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Shuffle Example Prompts
Create a line plot that displays multiple sets of x and y data
Roll two six-sided dice 1000 times and plot the sum of each roll
Solve the linear equations with coefficients A = [2, 4; 1, 3] and constants B = [8; 5] and display the results
ss3.m
Ask Copilot
Validate generated output before use.
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(iv)


Editor
ss3.m *
/MATLAB Drive/ss3.m
plot(xMeters / 1000, receivedPower1dBm, ...
'b', 'LineWidth', 1.5);
hold on;
plot(xMeters / 1000, receivedPower2dBm, ...
'r', 'LineWidth', 1.5);
xline(handoffPositionM / 1000, ...
'--k', 'Estimated Handoff');
xlabel('Mobile User Position (km)');
ylabel('Received Power (dBm)');
title('Received Signal Strength Between Adjacent Cells');
legend('Base Station 1', 'Base Station 2', ...
'Location', 'best');
grid on;
hold off;
drawnow;
Estimated Handoff Position: 433.0 m from Base Station 1
Received Signal Strength Between Adjacent Cells
Base Station 1
Base Station 2
Estimated Handoff
Received Power (dBm)
Mobile User Position (km)
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ss3.m
Ask Copilot
Validate generated output before use.
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